

Financial Management and Valuation: Class 1

SBWL Finance

Winter Term 2025-26

Please upload your solutions and indicate which Mini-Cases (MCs) you have solved before **December 8, 2025 at 10pm**. To do so, go to the assignment section on Canvas and choose “Unit 1 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions file upload will be considered, your last complete attempt counts.**

You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solutions consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism.

Exercise 1.1 There is a two-year bond with a face value of \$100 and with a coupon of 3% p.a., which is paid out semi-annually. The money market offers the following **linear** interest rates.

Maturity	Interest Rate p.a.
1M	3.5%
3M	3.6%
6M	3.7%
9M	3.9%
12M	4.0%
14M	4.3%
18M	4.5%
24M	5.2%

- (a) Write out the cashflows of the bond. Provide the cashflows of a tracking portfolio. Calculate the present value of the bond using a tracking portfolio. Refrain from rounding until the last step. Assume there is no arbitrage.
- (b) You are given that the price of another bond which has the same cashflows is trading at \$95.5. Is arbitrage possible?

Exercise 1.2 Consider the following **annually compounded** interest rates for different maturities at the end of each month between November 2025 and March 2026. These interest rates are used to price U.S. treasury bills.

	30/11/2025	31/12/2025	31/01/2026	28/02/2026	31/03/2026
3 month	3.5	3.3	3.7	3.5	3.4
6 month	3.7	3.6	3.8	3.6	3.3
9 month	3.9	3.7	3.9	3.3	3.1
12 month	4.2	3.8	4.1	3.2	2.9

- (a) Assume it is November 30th 2025 and you want to invest into a zero coupon 6 month treasury bill with face value 1000\$. The 6 month treasury bill is issued exactly on that day, and will mature six months later. What is the price you need to pay?
- (b) It is now February 28th 2026 (3 months have passed) and you wish to sell your investment. How much will you receive from your sale? What is the profit from your investment?
- (c) Do the same exercise for a 1 year treasury bill issued on November 30th 2025.

Exercise 1.3 You are given the three-year discount factor of 0.935.

- (a) Calculate the linear rate (p.a.).
- (b) Calculate the annual compounded rate (p.a.).
- (c) Calculate the period rate.
- (d) Is there a difference in the rates? Why or why not?

Exercise 1.4

- (a) Describe the cash flows associated with a forward contract, both for long and short position.
- (b) Write down the tracking portfolio for a short position in a forward on a stock.
- (c) What is the difference between the profit and the payoff from a forward contract? Explain why.

Exercise 1.5

You are interested in a forward contract on a non-dividend paying stock for a purchase of the underlying stock in three years ($t = 3$). The current stock price is $S_0 = 65$.

- (a) Assume an annually compounded risk-free rate of 4%, which forward price are you willing to agree on today?
- (b) After two years, i.e. at $t = 2$, the share price is $S_1 = 90$. What is the forward price of a fair forward contract to purchase one share at $t = 3$? The risk-free rate is unchanged.
- (c) At $t = 2$, what is the value of the forward contract which you bought at $t = 0$?

Exercise 1.6

Assume you are a U.S. investor, and that the exchange rate S_0 is 0.8 (number of € in exchange for 1\$). The domestic interest rate is 5% per period and the foreign (€) interest rate is 2% per period. You want to enter an FX forward to sell \$ (buy €) in one period. What is the no arbitrage forward rate? Show how to obtain it through a replicating portfolio.